Semantic Reactor Theory

Governance-Native Architecture for High-Authority Semantic Transformation Systems

SRT is descriptive before prescriptive: it measures governance legibility, not moral virtue.

Author
Adam Ableman
Constitutional Computing
Contact: ableman.research@gmail.com

Abstract

High-authority systems — courts, platforms, regulatory agencies, AI models, operating systems, financial instruments — share a common structure: they receive symbolic inputs, transform them under an authority structure, and produce binding effects on behavior, identity, legitimacy, access, belief, or institutional reality.

> **Definition (Semantic Reactor).**
> A system S is a semantic reactor if and only if: (1) S receives symbolic or semantic inputs, (2) S transforms those inputs under an authority structure that determines admissibility, and (3) S produces binding effects — effects externally consequential even when the target resists them.

We introduce **Semantic Reactor Theory (SRT)**, a governance-native framework modeling any semantic reactor as the tuple $S = (\mathcal{W}, \mathcal{G}, \mathcal{E})$ — Steward, Governor, Executor — with a governance trace $\tau = (EP, GR, ER)$ and an awareness vector $SRSA(S) \in \{0,1\}^{10}$. The SRSA score is $\|SRSA(S)\|_1 = \sum_{i=1}^{10} SRSA_i(S)$.

We identify **system flattening** $\varphi(S) \in [0,1]$ as a formal architectural error: a reactor with $\varphi(S) > 0$ cannot generate a governance trace τ that is decomposable from external artifacts alone — making its binding effects not structurally auditable from external artifacts.

SRT is descriptive before prescriptive: it measures governance legibility, not moral virtue. A system with $\|SRSA(S)\|_1 = 10$ is structurally transparent; it is not thereby morally correct.

---

1. The Governance Gap and Semantic Execution Density

High-authority systems exhibit a persistent structural pathology — one whose urgency is increasing.

> **Definition (Semantic Execution Density).** The semantic execution density of a society or system is the rate at which binding semantic transformations are performed per unit time.

Semantic execution density has grown dramatically in the last two decades. Platform algorithms perform billions of content governance decisions daily. Credit systems make binding access determinations for hundreds of millions of people. AI models increasingly mediate institutional decisions affecting employment, healthcare, legal process, and financial access.

As semantic execution density increases, the cost of low SRSA compounds. A court system with undecomposed authority might produce unjust outcomes affecting thousands of people annually; a content moderation algorithm with the same structural deficiency affects millions of outcomes daily; an AI system mediating institutional decisions at scale with merged decision/effect surfaces operates as undecomposed authority at civilizational throughput.

The volume of consequential semantic transformation has outrun the governance structures available to manage it. The central diagnosis of this paper: modern civilization increasingly operates high-throughput semantic reactors without decomposable governance traces. SRT provides the diagnostic framework and architectural vocabulary for closing that gap.

---

2. The Semantic Reactor Class

Binding != influence. A newspaper influences readers; a court judgment binds the parties even when they resist — failure to comply carries enforceable consequences. A recommendation algorithm influences behavior; a platform's content removal decision binds the affected party regardless of agreement. The discriminating criterion is external consequentiality under resistance.

Negative example. A pocket calculator is not a semantic reactor. A tax authority system using the same arithmetic to produce assessments binding on individual financial obligations is a semantic reactor — its outputs constrain enforceable state regardless of whether the recipient agrees. The binding criterion is relational and institutional, not intrinsic to the computation.

Binding types:

- **Binding type:** Physical
  Mechanism: State transition in hardware or infrastructure
  Example reactor: Operating system

- **Binding type:** Institutional
  Mechanism: Enforceable consequential determination
  Example reactor: Court, regulatory agency
- **Binding type:** Social
  Mechanism: Persistent alteration of access or visibility
  Example reactor: Platform content moderation
- **Binding type:** Economic
  Mechanism: Constrained financial state with enforceable consequences
  Example reactor: Financial instrument, credit system
- **Binding type:** Cognitive
  Mechanism: Meaning alteration coupled to dependency, authority, or constrained downstream behavior
  Example reactor: Curriculum authority, diagnostic system

Cognitive binding requires precision: belief alteration is binding when coupled to an authority relationship or to constrained downstream behavior such that acting contrary carries enforceable consequences. A compulsory curriculum with assessment authority is cognitively binding; a documentary is not.

Low SRSA means absent, undeclared, or unmeasured governance structure — not absent governance properties. Every semantic reactor has governance properties; low SRSA means those properties are dark.

Substrates. Semantic reactors may exist on biological, institutional, computational, and hybrid substrates. The three criteria are substrate-independent. Governance itself is substrate-independent: role separation, artifact production, and closed feedback loops constitute good governance regardless of whether the substrate is computational, institutional, or physical.

---

3. A Structural Analogy: The Engine Governor

A small-engine governor system is structurally analogous to the SR three-role model. The lawnmower engine is not a semantic reactor — it does not transform meaning under institutional authority into binding social effects. It is a structural analogy only.

- **Engine component:** Carburetor + sensing
  SR role: Steward ($\mathcal{W}$)
  Function: Maintains awareness; produces current state; does not decide
- **Engine component:** Governor spring + flyweight
  SR role: Governor ($\mathcal{G}$)
  Function: Applies setpoint rule to sensed state; controls throttle
- **Engine component:** Throttle + combustion
  SR role: Executor ($\mathcal{E}$)

Function: Produces effects; exhausts byproducts back to sensing layer

The governor mechanism is not in the power path. Collapse it — connect carburetor directly to throttle — and the engine races uncontrollably or stalls under load: not component failure, but a wrong control architecture. Semantic reactors exhibit the same failure under structural flattening.

---

4. The Formal Reactor Model

The reactor tuple:

$$S = (\mathcal{W}, \mathcal{G}, \mathcal{E})$$

where $\mathcal{W}$ is the Steward (epistemic position maintenance), $\mathcal{G}$ is the Governor (admissibility and authority decision under a declared authority structure), and $\mathcal{E}$ is the Executor (effect production and receipt). Role separation requires distinct interfaces, distinct authority scopes, and distinct record types — not labeled partitions of a unified process.

The governance trace and decomposability:

$$\tau = (EP, GR, ER)$$

where EP (EpistemicPosition) is produced by $\mathcal{W}$ and consumed by $\mathcal{G}$; GR (GovernanceReceipt) is produced by $\mathcal{G}$ and consumed by $\mathcal{E}$; and ER (ExecutionReceipt) is produced by $\mathcal{E}$ and fed back to $\mathcal{W}$.

> **Definition (Decomposable Governance Trace).** A governance trace $\tau = (EP, GR, ER)$ is decomposable if and only if EP, GR, and ER are distinct artifacts independently inspectable by an outside observer without privileged access to the process that produced them.

Most institutional records — logs, justifications, narratives, case notes — are not decomposable governance traces. They are outputs of a unified process from which EP, GR, and ER cannot be separately reconstructed. The distinction between a narrative trace and a decomposable governance trace is structural, not a question of record-keeping quality.

The governance loop:

$$\mathcal{W} \xrightarrow{EP} \mathcal{G} \xrightarrow{GR} \mathcal{E} \xrightarrow{ER} \mathcal{W}$$

The SRSA measurement vector (binary introduction; continuous relaxation in Limitations):

$$SRSA(S) \in \{0,1\}^{10}, \quad \|SRSA(S)\|_1 = \sum_{i=1}^{10} SRSA_i(S)$$

where $SRSA_i(S) = 1$ if axis i is instrumented, declared, and measurable, and 0 otherwise. Each component is a measurable structural property — a determination of whether a specific artifact type exists and is readable, not a subjective assessment.

The flattening degree:

$$\varphi(S) \in [0,1], \quad \varphi(S) = 1 - \text{role_separation_degree}(S)$$

where $\varphi(S) = 0$ denotes full role separation and $\varphi(S) = 1$ denotes full flattening. Flattening may be structural (formal roles merged), procedural (roles separated on paper but not in practice), or epistemic (roles formally separated but independence collapsed by authority routing or discovery constraints). A formally separated court whose judges receive pre-briefed summaries that embed the governance decision is epistemically flattened at the $\mathcal{W}$-$\mathcal{G}$ boundary regardless of formal role declarations.

Governance closure:

> **Definition (Governance-Closed Reactor).** A semantic reactor S is governance-closed if and only if every binding effect produced by S participates in a decomposable governance trace.

---

5. Governance-Native Architecture

> **Definition (Governance-Native Architecture).** An architecture is governance-native if production of τ is structurally required for effects to occur — not merely for effects to be recorded after the fact.

Overlay governance	Governance-native architecture
Policies added after execution is designed	Governance constitutive of execution from the start
Audit after effect	Receipt-before-effect
External compliance checking	Internal admissibility evaluation
Runtime patching of governance gaps	Architectural role separation
Governance as constraint on behavior	Governance as structure of behavior

The Governance Event Principle. An effect without a prior GR is an epistemic non-event: it happened physically, but it did not happen in the

governed sense. The governance system cannot see it because it produced no artifact. This is the constitutive claim distinguishing governance-native from overlay architecture. In an overlay system, an ungoverned effect is a policy violation to be detected after the fact. In a governance-native system, an ungoverned effect cannot advance the lineage — it is invisible to $\mathcal{W}$ on the next pass.

Decomposable governance traces compress the semantic explanation burden of authority. Without decomposition, observers must reconstruct intent psychologically or socially. With decomposition, governance becomes mechanically inspectable: the question "on what basis was this binding effect produced?" has a direct, artifact-grounded answer.

---

6. System Flattening as Formal Architectural Error

> **Definition (System Flattening).** System flattening occurs when two or more of $\mathcal{W}$, $\mathcal{G}$, $\mathcal{E}$ are performed by the same entity such that τ is no longer decomposable from external artifacts.

System flattening is not a moral pathology. It is a formal architectural error with a specific, auditable consequence.

Structural Result 1 (Non-Decomposability of Flattened Governance). A semantic reactor with $\varphi(S) > 0$ cannot generate a governance trace τ that is decomposable from external artifacts alone.

Proof. Decomposability requires EP, GR, and ER to be independently inspectable. When $\mathcal{W}$ and $\mathcal{G}$ are merged, epistemic synthesis and constitutional application occur in a single process; EP is consumed internally and not produced as a distinct external output. An observer with access only to GR cannot reconstruct the EP from which the decision was derived without privileged access to the internal state. □

The judge problem. A judge performing both evidence synthesis ($\mathcal{W}$) and constitutional application ($\mathcal{G}$) produces a judgment from which EP and GR cannot be separated by an outside observer. Appellate systems distinguish fact review from law review precisely because they are post-hoc structural separation imposed on a partially flattened process.

The algorithm problem. A content moderation algorithm with $\mathcal{W}$, $\mathcal{G}$, and $\mathcal{E}$ merged produces no EP, no GR, and no ER feedback. The output is the only artifact; the governance process is not reconstructible from it.

Structural Result 2 (Flattening Degrades Under Scale). A semantic reactor with $\varphi(S) > 0$ produces governance quality dependent on the capability of the unified role. As complexity or adversarial pressure increases beyond that capability, governance quality degrades. In a separated system, error at any handoff is detectable by the receiving role from its input artifact.

Proof sketch. In a separated system, EP errors are detectable by $\mathcal{G}$; GR errors by $\mathcal{E}$; ER errors by $\mathcal{W}$ on the next pass. Error detection scales with distinct artifact handoffs, not with individual capability. □

---

7. Semantic Reactor Situational Awareness (SRSA)

SRSA is a measure of governance legibility — not runtime observability. The distinction is foundational:

Classical observability	SRSA
Runtime state and system metrics	Governance state across all ten axes
Logs and traces	Admissibility rules and invariant declarations
Operational debugging	Governance completeness
Performance visibility	Incentive geometry (designed and emergent)
"Is the system working?"	"Is the system legitimate?"

A fully observable system with all metrics green and all logs retained may have $\|SRSA(S)\|_1 = 2$ if its authority topology is undeclared, its incentive geometry unmapped, its admissibility conditions implicit, and its legitimacy basis unexamined.

Every semantic reactor possesses awareness state across ten dimensions whether or not that state is declared. The ten axes:

- **Axis:** ESA
  Name: Epistemic Awareness
  Core question: What does the system know, verify, and declare unknown?
- **Axis:** SSA
  Name: Semantic Awareness
  Core question: What meanings does it transform, under what authority structure?
- **Axis:** ISA
  Name: Incentive Awareness
  Core question: What behavior does it actually reward — designed and emergent?
- **Axis:** ASA
  Name: Authority Awareness
  Core question: Who can define, interpret, modify, enforce, or refuse?

- **Axis:** CSA
  Name: Causal Awareness
  Core question: What caused this effect, through which pathway?
- **Axis:** PSA
  Name: Provenance Awareness
  Core question: Where did this artifact, rule, or authority originate?
- **Axis:** AdSA
  Name: Admissibility Awareness
  Core question: What may enter without violating declared invariants?
- **Axis:** RSA
  Name: Risk Awareness
  Core question: What failure modes and exploit paths are currently active?
- **Axis:** FSA
  Name: Fidelity Awareness
  Core question: Does the representation still match the thing represented?
- **Axis:** LSA
  Name: Legitimacy Awareness
  Core question: Why is this system entitled to bind anyone? When does that fail?

Orthogonal but coupled. Each axis is independently variable — $ESA = 1$ is compatible with $ISA = 0$ — while remaining dynamically interacting. FSA drift may degrade ESA if the epistemic position is built from representations no longer matching reality. This is why ASA and LSA are distinct axes: a fully declared authority topology ($ASA = 1$) with an entirely unclear legitimacy basis ($LSA = 0$) is a different governance failure from the reverse.

SRSA does not imply moral correctness. A dictatorship could have $\|SRSA(S)\|_1 = 10$. A system with full SRSA producing unjust outcomes is a legible injustice. A system with $\|SRSA(S)\|_1 = 2$ producing just outcomes produces justice through opacity. SRT addresses the first problem; the second is a question of values.

Structural Completeness Proposition. Every semantic reactor possesses instrumentable SRSA state across all ten axes, regardless of declaration.

Proof. A semantic reactor transforms inputs under an authority structure into externally consequential outputs. This constitutes, structurally: an epistemic position (ESA), a semantic transformation structure (SSA), an incentive gradient (ISA), an authority topology (ASA), a causal structure (CSA), provenance of rules and artifacts (PSA), admissibility conditions (AdSA), failure modes (RSA), representation fidelity (FSA), and a legitimacy basis (LSA). These properties exist whether or not declared. □

---

8. The ISA/SSA Gap as Primary Drift Surface

> **Definition (ISA/SSA Gap).** The ISA/SSA gap is the divergence between a reactor's declared transformation purpose (SSA) and the behaviors its incentive structure actually rewards (ISA), including incentives not designed by governance.

The ISA axis captures designed and emergent incentives. A court's incentive structure includes career advancement dynamics, caseload pressure, and political appointment considerations — none appearing in the constitutional codex. A platform's includes advertising revenue optimization. An AI model's includes training reward signals and deployment pressures.

Structural Result 3 (ISA/SSA Gap Growth Under Opacity). In a system with low ASA and low PSA, the ISA/SSA gap grows monotonically over time as undeclared authority surfaces accumulate, rules acquire informal interpretations, and incentive gradients reinforce behaviors serving authority holders rather than declared mission.

This is the mechanism of institutional capture — structural accumulation of undecomposed authority and incentive misalignment in low-SRSA systems, independent of individual intentions.

---

9. The Governance Loop and a Concrete Trace

> **Definition (Closed Governance Loop).** A semantic reactor has a closed governance loop if and only if: (1) every $\mathcal{W}$-to-$\mathcal{G}$ handoff produces a distinct EP, (2) every $\mathcal{G}$-to-$\mathcal{E}$ handoff produces a distinct GR, (3) every ER is available to $\mathcal{W}$ on the next pass, and (4) $\mathcal{G}$ receives only validated EP — not raw evidence directly.

Condition (4) is the architectural invariant preventing $\mathcal{W}$-$\mathcal{G}$ flattening.

Structural Result 4 (Open Loop Implies Non-Decomposability). A semantic reactor with an open governance loop is not structurally auditable from external artifacts in the same sense as a flattened reactor. Each missing artifact eliminates the corresponding decomposability condition.

Concrete governance trace. A content moderation system with full role separation:

Steward pass (EP). $\mathcal{W}$ reads the content and receipt history, producing: $EP = \{$known: matches pattern X, confidence 0.87; unknown: authorial intent; dark: off-platform context$\}$. This is a distinct artifact — not the governance decision.

Governor pass (GR). $\mathcal{G}$ receives EP and applies the admissibility codex: pattern X at confidence ≥ 0.8 without exception $\rightarrow$ REMOVE admissible. It produces: $GR = \{$decision: REMOVE; rule: §3.2; EP_hash$\}$.

Executor pass (ER). $\mathcal{E}$ receives GR, removes the content, and produces: $ER =$ {action: removed; GR_hash; timestamp}.

Feedback. ER is available to $\mathcal{W}$ on the next pass. An outside observer can reconstruct what the system knew (EP), what rule was applied (GR), and what happened (ER). The governance trace is decomposable. The authority is no longer undecomposed.

---

10. Constitutional Consensus: A Prior Legitimacy Layer

Constitutional Consensus is a prior legitimacy layer — not a replacement for Raft or Paxos, but a prior layer that establishes shared governance identity before event ordering matters. Classical consensus then orders events within the established legitimacy substrate.

> **Definition (Constitutional Consensus).** Constitutional Consensus is a distributed coordination model in which participants first share a content-addressed governing constitution c, establishing session_constitution_hash $= H(\text{canon}(c))$, and then independently verify actions, quorum, authority, ordering, and divergence against c through receipted artifacts.

Constitutional Consensus relocates the coordination burden from synchronization-time to constitution-adoption-time. Classical consensus performs agreement at the moment events occur; Constitutional Consensus performs agreement when participants adopt the same constitution. Runtime activity then consists of verification, not negotiation.

- **Property:** Primary question
  Classical (Raft/Paxos/PBFT): What happened and in what order?
  Constitutional Consensus: Under what authority did this happen?
- **Property:** Agreement mechanism
  Classical (Raft/Paxos/PBFT): Runtime voting / leader election
  Constitutional Consensus: A priori constitutional identity
- **Property:** Validity rules
  Classical (Raft/Paxos/PBFT): External to consensus
  Constitutional Consensus: Internal to the constitution
- **Property:** Divergence
  Classical (Raft/Paxos/PBFT): System failure
  Constitutional Consensus: Typed governance event
- **Property:** Quorum
  Classical (Raft/Paxos/PBFT): Transient voting round
  Constitutional Consensus: Content-addressed evidence

- **Property:** Coordination burden
  Classical (Raft/Paxos/PBFT): Per-event synchronization
  Constitutional Consensus: Constitution-adoption-time

Structural Result 5 (A Priori Agreement). Participants sharing session_constitution_hash are in constitutional agreement by definition. The handshake verifies this agreement; it does not produce it.

---

11. Formal Properties

Lemma (Receipt Non-Repudiation). Under collision-resistant H and injective canon$(\cdot)$, the receipt $r = (step_hash, law_hash, outcome, payload_digest)$ uniquely and non-repudiably binds a specific step to a specific law evaluation and outcome.

Proof. Each component hash is unique under H and canon$(\cdot)$ by collision resistance. Modifying any component produces a detectable mismatch. Under second-preimage resistance, computing a different step with the same $step_hash$ is infeasible. □

Structural Result 6 (Policy Non-Retroactivity). Under law_hash binding at step creation time, amending a law from L_v to L_{v+1} does not alter the admissibility determination of any step s with $s.law_hash = H(\text{canon}(L_v))$.

Proof. $s.law_hash$ is immutable at construction. Evaluation under L_{v+1} is blocked because $s.law_hash \neq H(\text{canon}(L_{v+1}))$. □

---

12. Software as First Tractable Substrate

Software systems are the first substrate on which full SRSA instrumentation is tractable. The Semantic Defragmentation Analysis (SFA) methodology has been applied to seventy-nine open-source systems across eleven domains, producing structured thirteen-phase analyses identifying authority sources, admissibility conditions, incentive gradients, transformation paths, and receipt surfaces. The same structural pathologies appeared consistently — undecomposed authority, undeclared capability surfaces, absent receipt emission, projection divergence — until their structural unity across domains became unmistakable.

Software tractability rests on three properties: epistemic state can be instrumented mechanically through receipt chains; authority topology can be declared formally in a constitutional codex; and effects can be made constitutive through the receipt-before-effect discipline. The methodology transfers to institutional substrates — harder in instrumentation, not different in structural kind.

---

13. Limitations

SRT measures governance structure; intervention recommendations follow from measurement.

High SRSA does not imply good governance — a legible injustice is still an injustice. Instrumentable SRSA state does not mean easy measurement — institutional ISA, legitimacy chains, and authority topology require significant analytical work. SRT does not claim total formalizability — institutions contain genuine ambiguity and irreducible discretion. The binary $SRSA(S) \in \{0,1\}^{10}$ simplifies; the continuous relaxation $SRSA_i(S) \in [0,1]$ and the corresponding continuous $\varphi(S)$ are the empirically appropriate models.

---

14. Related Work

Constitutional and institutional theory has examined separation of powers, checks and balances, and judicial independence as governance mechanisms. Montesquieu's trias politica is an early statement of the role separation SRT formalizes. SRT contributes the formal tuple $S = (\mathcal{W}, \mathcal{G}, \mathcal{E})$, the governance trace τ, the SRSA vector, and $\varphi(S)$ as a formal architectural error class with instrumentable structure.

Distributed systems theory — Lamport et al., Oki and Liskov's Viewstamped Replication, Paxos, and Raft — addresses event ordering under the assumption that validity rules are external. Constitutional Consensus addresses the prior question: how participants establish shared validity rules before events occur.

Platform governance literature (Zittrain, Gillespie, Balkin) identifies governance functions of algorithmic systems without a unified structural framework. The semantic reactor definition, flattening diagnosis, and SRSA vector provide a formal spine for those architectural observations.

Prior uses of "semantic reactor." The term has prior independent uses that this paper acknowledges. General Semantics literature — notably in the tradition of Korzybski — used "semantic reactor" to describe human beings as organisms that abstract, interpret, and react semantically to symbolic inputs. This usage predates and is orthogonal to the present framework: it refers to a biological cognitive property rather than an architectural class. A Google Research tool named "Semantic Reactor" provides a natural language interface for ranking and sorting text via semantic similarity models — a product name with no overlap with the present theoretical sense. A small body of work in AI systems uses "semantic reactor" informally to describe AI architectures that process meaning. SRT adopts the term as a formal technical class because it captures the essential structure — semantic input, authority-mediated transformation, binding output — more precisely than available alternatives, while acknowledging these prior independent coinages.

AI safety and alignment research, including Constitutional AI, addresses related questions but does not provide the $(\mathcal{W}, \mathcal{G}, \mathcal{E})$ separation or $SRSA$ measurement framework. SRT provides the architectural substrate in which such rules serve as $\mathcal{G}$'s authority source.

---

15. Future Work

Byzantine fault tolerance under Constitutional Consensus. Structural Result 5 does not bound equivocating participants. Formal tolerance bounds, liveness under partition, and constitutional fork governance require development.

SRSA measurement protocols for institutional substrates. The thirteen-phase SFA methodology requires extension to courts, regulatory agencies, and platforms via domain-specific intermediate representation development.

ISA/SSA gap formalization. Formal characterization of gap growth rates, detection thresholds, and intervention conditions requires development.

Reproducible SRSA assessments. SRT becomes empirically real when independent investigators produce convergent $SRSA(S)$ scores. Scoring rubrics and inter-rater reliability protocols are the near-term empirical frontier.

Continuous SRSA and flattening metrics. The relationship between partial $\varphi(S)$ and governance decomposability quality requires formal characterization.

Optimal decomposition regime. Decomposition improves governance decomposability but introduces coordination overhead, latency, and artifact inflation. The relationship between decomposition degree and governance quality — and the conditions under which further decomposition becomes counterproductive — requires formal treatment.

Reactor coupling. The next analytical frontier is coupled reactor systems: what happens when semantic reactors interact recursively? Platforms, media ecosystems, governments, AI systems, and markets are analyzable as coupled semantic reactors whose SRSA properties co-evolve. Emergent ISA gradients, cascading legitimacy failure, and governance amplification at civilization scale become analyzable phenomena under reactor coupling theory.

Semantic metabolism. The governance loop describes a metabolically coherent process: intake, transformation under admissibility, effect production, receipt as exhaust, and feedback into the awareness surface. Formalizing semantic metabolism would provide vocabulary for analyzing long-horizon reactor stability and governance regulation of semantic throughput.

---

16. Conclusion

Semantic Reactor Theory establishes:

Any system producing binding effects under an authority structure is a semantic reactor possessing an instrumentable $SRSA(S) \in \{0,1\}^{10}$ state vector, whether or not declared.

System flattening $\varphi(S) > 0$ is a formal architectural error. A reactor with $\varphi(S) > 0$ cannot generate a governance trace $\tau = (EP, GR, ER)$ that is decomposable from external artifacts alone. The structural production of apparent arbitrariness in institutional authority — the visceral sense that power is exercised without decomposable justification — is a consequence of this architectural condition, not of individual malice or incompetence.

The ISA/SSA gap is the primary drift surface for institutional capture, growing under opacity independently of individual intentions.

Constitutional Consensus is a prior legitimacy layer establishing shared governance identity at constitution-adoption-time, beneath classical fault-tolerant distributed coordination.

Semantic reactors can be instrumented. Flattening can be detected and measured as $\varphi(S)$. $\|SRSA(S)\|_1$ can be scored. Governance decomposability can be structurally improved. The foundational architectural primitives — AOK, MCIR, SRSA methodology, Constitutional Consensus — now exist as emerging tools; their application to the systems that bind human life is the work ahead.

SRT measures governance legibility and structural decomposability — preconditions for accountability, not substitutes for it.

Acknowledgments

The author extends thanks to the research tradition in constitutional and institutional design — from Montesquieu through Madison through modern platform governance scholarship — whose insights this work attempts to formalize. The development of SRSA was grounded in repeated practical application of the Semantic Defragmentation Analysis methodology across seventy-nine open-source systems spanning eleven domains, providing empirical evidence that the same structural governance pathologies appear consistently across substrates.

Corpus-Wide Reference Positioning

This paper argues that high-authority systems — courts, platforms, regulatory agencies, AI models, operating systems, financial instruments — share a common structure: they receive symbolic inputs, transform them under an authority structure, and produce binding effects on behavior, identity, legitimacy, access, belief, or institutional reality. Its nearest external comparison fields are semantic,

formal, governance. The sources in [E1], [E2], [E3], [E4], [E5], [E6], [E7] are cited as predecessor literature, adjacent standards, or counterpressure - not as evidence that they caused the Ableman framework or that they validate its novel claims. The comparison isolates where established work already supplies provenance, formal assurance, institutional constraint, empirical method, or engineering practice, and where this paper adds a distinct claim about admissibility, authority, semantic custody, or governed realization.

Within the Ableman corpus, the highest-priority current cross-links are [A1], [A2], [A3], [A4], [A5], [A6], [A7]. These are links among separately custodied works, not silent identity mergers. Their role is to expose mutual dependencies, terminology collisions, and places where a reader should traverse the corpus before treating this paper as self-sufficient.

Reference-pass counterevidence and limit search

The pass applied the following pressure tests: Formal vocabularies and graph links make commitments explicit but cannot by themselves establish correct identity, shared interpretation, or lossless translation between contexts. Formal assurance remains relative to the encoded model, trusted base, and refinement chain; proof does not remove specification error or unmodeled environmental behavior. A written constitution can legitimate domination as easily as constrain it unless amendment, participation, enforcement, and appeal remain effective in practice. These limits are not generic disclaimers: they identify the assumptions that a later empirical, formal, or implementation review must try to break.

Reference-pass influence and novelty boundary

No newly added external source is asserted to be a documented authorial influence unless the predecessor source already said so. `Predecessor` means earlier relevant work;

`adjacent` means a useful comparison; `standard` means an interoperability or assurance baseline; and `counterevidence` marks work that narrows or pressures a claim. Novelty therefore remains a reviewable, claim-level question rather than an inference from a low similarity score or an absent citation.

References

External scholarship, standards, and prior art

[E1] Richard Cyganiak, David Wood, and Markus Lanthaler, eds. *RDF 1.1 Concepts and Abstract Syntax.* W3C Recommendation, 25 February 2014. https://www.w3.org/TR/rdf11-concepts/ **Relationship:** Adjacent graph data model and identity baseline.

[E2] Alistair Miles and Sean Bechhofer, eds. *SKOS Simple Knowledge Organization System Reference.* W3C Recommendation, 18 August 2009.

https://www.w3.org/TR/skos-reference **Relationship:** Adjacent standard for concepts, labels, schemes, notes, and semantic mappings.

[**E3**] W3C OWL Working Group. *OWL 2 Web Ontology Language Document Overview*, Second Edition. W3C Recommendation, 11 December 2012. https://www.w3.org/TR/owl2-overview/ **Relationship:** Adjacent formal ontology standard used to compare declared semantics and inference commitments.

[**E4**] Mark D. Wilkinson et al. "The FAIR Guiding Principles for Scientific Data Management and Stewardship." *Scientific Data* 3, 2016, 160018. DOI: 10.1038/sdata.2016.18. https://doi.org/10.1038/sdata.2016.18 **Relationship:** Predecessor stewardship principles for findability, accessibility, interoperability, and reuse.

[**E5**] Peter Gardenfors. *Conceptual Spaces: The Geometry of Thought*. MIT Press, 2000. mitpress.mit.edu source **Relationship:** Predecessor geometric treatment of concepts; cited as comparison, not as validation of the paper's manifold terminology.

[**E6**] George C. Necula. "Proof-Carrying Code." *POPL '97*, 1997, pp. 106-119. DOI: 10.1145/263699.263712. https://doi.org/10.1145/263699.263712 **Relationship:** Predecessor for binding machine-checkable evidence to executable artifacts; the present paper may demand broader semantic or constitutional obligations.

[**E7**] Xavier Leroy. "Formal Verification of a Realistic Compiler." *Communications of the ACM* 52(7), 2009, pp. 107-115. DOI: 10.1145/1538788.1538814. https://doi.org/10.1145/1538788.1538814 **Relationship:** Predecessor verified-compilation result used to distinguish proven translation preservation from broader governance claims.

Ableman corpus

[**A1**] Adam Ableman. *Meaning as Binding*. Ableman Corpus current edition; work identity
`AMCS-LOCAL-WORK-5F976917F0B283EA`;
source SHA-256
`A470D1A0B9ED7C3CEAB6BB88BE4B642A2BA36427447F787764329FD574DA0C97`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A2**] Adam Ableman. *Governed Meaning Base (GMB)*. Ableman Corpus current edition; work identity
`AMCS-LOCAL-WORK-D95CCC4FB7B58BFB`;
source SHA-256
`A3B62E50F8BEE125FDAE0D38C2839AFD1CCC138296268BF93217481E38AA48EA`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A3**] Adam Ableman. *Semantic Manifold Topology.* Ableman Corpus current edition; work identity `PAPER.PUB.20257244`;
source SHA-256
`B8F28CC652A73A3EE4CF6FD0E23C1ABC922A6660319B2FB4CC1CCAAA606EC64A`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A4**] Adam Ableman. *Global Term Key: Identity Primitives in Governed Knowledge Systems.* Ableman Corpus current edition; work identity
`AMCS-LOCAL-WORK-358AC09F5A13587E`;
source SHA-256
`AFB29CFCB4B58A78B79709AD453CEE9088CCB2AE40F30A70260F1B54F5160579`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A5**] Adam Ableman. *Semantic Legitimacy Before Context.* Ableman Corpus current edition; work identity
`AMCS-LOCAL-WORK-091E34BA93992CE4`;
source SHA-256
`DEB7626CE1AB54F4B541347DCC3C4E14DC4F682230ACCCE1A93D1D536A50C44C`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A6**] Adam Ableman. *MCIR: A Canonical Intermediate Representation for Governed Transformation.* Ableman Corpus current edition; work identity `PAPER.PUB.21095303`;
source SHA-256
`EA129ECBA95F7B07CAA7B33A54F91F52913C94A14DA227EB9A06CFD083E4E5BF`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A7**] Adam Ableman. *Governed Semantic Compilation.* Ableman Corpus current edition; work identity `PAPER.PUB.18453681`;
source SHA-256
`5FF6C636550B04DB0E2F6F94C379BB75AA0BD8781E19B047D7D3BB6D49071AC2`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

Corpus source and companion formats: https://github.com/AblemanMosaic/ableman-corpus
License: CC-BY-ND-4.0

DOI: 10.5281/zenodo.20153069